Government of Goa

AGRICULTURAL CENSUS

REPORT ON MAIN CENSUS

2010-11

GOA

DIRECTORATE OF PLANNING,
STATISTICS & EVALUATION
PORVORIM-GOA

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PREFACE

Agricultural Census is conducted in the Country as a nationwide programme during every fifth year. Goa participates in this programme since Agricultural Census conducted with reference year 1970-71. The Directorate of Planning, Statistics & Evaluation has currently conducted the Agricultural Census (2010-11) which is 9th in the series with the Agricultural year 2010-11 as the reference period.

The data for Agricultural Census 2010-11 has been collected through the field inquiries from all the operational holdings from the selected villages. Since there is no primary reporting system in Goa, the data for the current census was collected through random sampling method. The 50 percent of the villages/wards in each Taluka were selected by using simple random sampling method and in each selected village/ward cent percent of the operational holdings were covered during the census. The selected villages were surveyed for collection of information on number and area of operational holdings according to size classes, land utilization, crop wise and source wise irrigated area, cropping pattern, tenancy, etc.

In order to ensure uniformity and comparability, the concepts and definitions adopted by the Government of India, Ministry of Agriculture, were used. This report gives a narrative account of the historical background, physical and demographic features, economy of the region, system of collection of agricultural statistics, etc.

The Directorate is grateful to a large number of cultivators who spared time to provide detailed information required for this Census. Similarly the Gram Sevaks, Talathis, Panchayat Secretaries, Investigators, Statistical Assistants, etc. who worked as Enumerators and Supervisors under the Census, the Mamlatdars, Research Assistants of this Directorate who functioned as Taluka Level Charge Officers and Additional Taluka Level Charge Officers also deserve the appreciation.

The Co-ordination Division of this Directorate headed by Joint Director Shri Vijay B. Saxena has shared the main burden in successfully conducting this Census right from the beginning. The preliminary draft of this report was prepared by Shri Deepak Pednekar, Statistical Officer, Shri. Laximan Amonkar, Research Assistant, with the help of Smt. Vandana Naik Panvelkar, Statistical Assistant, and Shri.James Cardozo, Investigator, who has taken the initiative to computerize the report. The valuable guidance of Shri Jagdish N. Shirodkar, Deputy Director, in preparing the report is equally appreciated.

It is hoped that the data published in this report would be useful for formulation and implementation of Agricultural Development Programmes.

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Director & Dy. Agricultural
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Porvorim – Goa
Dated : January, 2015

CHAPTER I

INTRODUCTION

1.1 Agriculture holds a key position and constitutes one of the predominant sector of the economy of Goa. The relevant information for micro level agricultural planning is gathered on an operational holding basis through the Agricultural Census. An "Operational Holding" for the purpose of Agricultural Census is defined as "all land which is used wholly or partly for agricultural production and is operated directly or managed as one technical unit by one person alone or with others without regard to the title, legal form, size or location. A holding may consist of one or more parcels of land provided they are within the same Talukas and form part of the same management and has the same means of production such as labour force, machinery and animals".

1.1.2 A World Agricultural Census was organized in 1924 by the International Institute of Agriculture, Rome. India participated in the World Agricultural Census in 1950 and in 1960. The data in these Censuses was collected through the National Sample Survey Organization.

1.1.3 The first Agricultural Census conducted in Goa (the then union territory alongwith Daman and Diu) with the reference year 1970-71 was followed by seven censuses held in quinquennial years of 1976-77, 1980-81, 1985-86, 1990-91, 1995-96, 2000-01 and 2005-06. The present census 2010-11 was the ninth one in the series of such censuses organized by the Ministry of Agriculture, Govt. of India, as a Centrally Sponsored Scheme.

1.2 Agricultural Census in the State of Goa

The erstwhile Union Territory of Goa, Daman & Diu participated in the 1970-71 Agricultural Census for the first time. The Census was conducted in accordance with the guidelines issued by the Ministry of Agriculture, Govt. of India, in order to ensure uniformity and comparability of census data between the States and Centre. For the Census of 1970-71 the data was collected through a special survey conducted for the purpose visiting each and every house. The method of retabulation of data from land records could not be adopted in the territory due to the absence of the cadastral survey and non-existence of the primary reporting system. The second census was conducted with 1976-77 as the reference year. The Third Agricultural Census was carried out as Central Scheme during 1980-81. The entire operation, unlike the last census was carried out in two phases namely Main Census and Input Survey. The fourth Agricultural Census and the subsequent censuses of fifth to eighth were conducted in the years 1985-86, 1990-91, 1995-96, 2000-01 and 2005-06.

The number of operational holdings and area operated as revealed under different censuses was as below:

Census	No. of operational holdings	% variation	Area operated (in Ha.)	% variation
1970-71	68961	---	92421	---
1976-77	72403	(+) 4.99	82667	(-) 10.55
1980-81	75537	(+) 4.33	89656	(+) 8.45
1985-86	75619	(+) 0.11	78129	(-) 12.86
1990-91	71922	(-) 4.89	66529	(-) 14.85
1995-96	70399	(-) 2.12	59022	(-) 11.28
2000-01	64080	(-) 8.98	53924	(-) 8.64
2005-06	52821	(-)17.57	60742	(+)12.64

1.2.1 The present Agricultural Census, ninth in the series was conducted in the state with reference year 2010-11. Data has been collected from each of the operational holding in the selected villages covering aspects such as size of holding, tenure and tenancy status, cropping pattern used, sources of irrigation used etc. during the census period.

The Secretary (Agriculture) was appointed as the Agricultural Census Commissioner and the Director of Planning, Statistics and Evaluation as Deputy Agricultural Census Commissioner. The Collector of North Goa and South Goa Districts were appointed as District Agricultural Census Officers and Mamlatdars of the respective Talukas as Charge Officers for smooth and efficient conduct of the Agricultural Census. The Deputy Directors, Statistical Officers, Research Assistants of Directorate of Planning, Statistics and Evaluation were appointed as Additional Charge Officers. Village Panchayat Secretaries, Talathis, Gram Sevaks were appointed as Enumerators. The Circle Inspectors, Extension Officers and Statistical Assistants of the Common Statistical Cadre were appointed as Supervisors.

CHAPTER II

STATE ECONOMY

2.1 The concluded Agricultural Census with the agricultural year 2010-11 as the reference year was the ninth in series of such censuses conducted so far in the State of Goa, as a part of National Programme. As in the last Census, this Census was also carried out in two phases called the "Main Census" with the agricultural year 2010-11 as the year of reference and the second phase called the "Input Survey" with the year 2011-12 as the reference period. Under the "Main Census" information pertaining to five broad characteristics, viz. i) number and area of operational holdings, ii) tenure and tenancy, iii) land utilization, iv) crop wise and source wise irrigated area, and v) cropping pattern, was collected.

2.1.2 Operational holding was the primary unit for collection of data so as to provide an integrated picture of the agricultural structure of the State which is a fundamental unit of decision making for formulation of plans for improving the individual / holding level cultivation especially of small and marginal farmers.

2.1.3 For 2010-11 Agricultural Census, the Ministry of Agriculture, Government of India, had suggested two methods. Under the first method, a complete enumeration was to be carried out in respect of items for which information was readily available from the revenue records directly by re-tabulation without making any further inquiry from the operational holder. This method, however, was not suitable for this state for lack of availability of up-to-date land records at village level from the cadastral survey as also for lack of data from primary reporting system which was not introduced. As such, the alternative method of 'enquiry on a sample basis' was adopted in this state.

2.1.4 For carrying out the survey, a two stage random sampling design with taluka as the stratum was adopted. The village was the first stage unit of sampling and operational holder was the second stage unit. From each taluka, 50 percent of the villages/wards were selected by simple random sampling method. In each selected village/ward cent per cent of operational holdings were covered for the main census.

2.2 Organization and administrative arrangements

A special cell comprising a Dy. Director, Investigator, L.D.C. cum typist and a Peon was created for the census purpose in the Directorate of Planning, Statistics and Evaluation. The Dy. Director of this cell was operational in charge of the census while the Director of the Directorate of Planning, Statistics and Evaluation was Dy. Agricultural Census Commissioner for the state who took overall responsibility of the census work. The Secretary (Agriculture), Govt. of Goa was appointed as the Agricultural Census Commissioner. The Taluka Mamlatdars were appointed as the Charge Officers in their respective talukas under the overall supervision of the Collectors who were appointed as District Census Charge Officers.

2.3 Training of personnel:

For the purpose of the main census, a two-tier training programme was drawn according to plan. First, the State level Training Seminar for the Charge Officers was held at Panaji. In the second phase, training to field workers was imparted in the training camps at taluka level.

2.3.2 A meeting of supervisors for discussing the various check points to be practiced while supervising and spot checking during the field work was also organized by the Directorate of Planning, Statistics and Evaluation.

2.4 Schedules and instructions:

The Directorate of Planning, Statistics and Evaluation drafted the various schedules and instruction for conducting the census on the basis of the model schedules provided by the Ministry of Agriculture, Government of India. During the training, all the schedules to be canvassed by the enumerators were discussed thoroughly.

2.4.2 In order to ensure proper direction and co-ordination of the Agricultural Census operations, a State Level Co-ordination Committee was constituted under the Chairmanship of the Commissioner and Secretary (Agriculture), the Collectors of North and South Goa, Director of Agriculture, Project Director of Rural Development Agency, Director of Computer Centre, representative of National Informatics Centre, Pune and representative of Ministry of Agriculture, Government of India being the Members and Director, Directorate of Planning, Statistics and Evaluation, being the Member Secretary. A State Level Conference was organized by the Director, Directorate of Planning, Statistics and Evaluation to discuss all the schedules and instructions and arrangements for training and other administrative methods.

2.4.3 In order to obtain realistic data, a press note was released through the Directorate of Information in the local dailies, publishing information about the Main Census of Agriculture, 2010-11, personnel engaged for the conduct of census, type of enquiry proposed during the visit to households etc. and requesting co-operation from the public in furnishing the required information correctly.

CHAPTER III

CONCEPTS, METHODOLOGY, SCOPE & COVERAGE

3.1 Physical features, Rainfall, Temperature, Crop-Season and Population characteristics:

Goa attained statehood on 30th May, 1987. It constitutes two Districts, North Goa & South Goa. It is located on the West Coast of the Indian Peninsula and between the coastal borders of Maharashtra and Karnataka.

3.1.2 The State of Goa is situated between the parallels $15^0 - 47' - 59"$ and $14^0 - 53' - 57"$ of latitude north and between meridians $73^0 - 40' - 54"$ and $74^0 - 20' - 11"$ of longitudes east of Greenwich. Its length from the extreme south to north measures 105 kms. and the width from east to west measures 60 kms. It covers an area of 3702 sq. kms. The altitude of the towns of this state range between 20 and 62 meters above the sea level.

3.1.3 The boundaries of this State are well defined in the north by the Tiracol River which separates the State from Maharashtra State in the east and south by Karnataka State while in west it is bounded by the Arabian Sea.

3.1.4 The State of Goa has a hilly terrain especially on its eastern side, where lies the southern end of the Sahyadri ranges. These mountains, after skirting a considerable portion of the northeastern and southeastern boundaries, branch off westward across the state with many spurs and ridges. The terrain is interspersed by a number of rivers flowing westwards, which provide a network of internal waterways. The important rivers worth mentioning are Mandovi, Zuari, Tiracol, Chapora and Betul which are navigable for a total length of 256 kms. The coast is full of creeks and estuaries formed by these rivers which provide a good shelter for fishing crafts. Estuaries of these rivers are rich in marine fauna. The 105 kms. long coastal line of Goa is endowed with some of the loveliest beaches in the world which have earned the fame of bearing idyllic beauty spots and are visited by the holiday makers and lovers of nature both domestic and foreign every year.

3.1.5 The physiographic characteristics are rich and varied consisting of verdant hills, forests, coconut groves and paddy fields. The most picturesque region is located in the lower basins of the rivers of Mandovi and the Zuari, which join to form a bay. There are about half a dozen inlets between the estuaries of these rivers, the largest of which is Goa from which the whole state appears to have derived its name. The port of Mormugao situated on the southern side of the Zuari is one of the best natural harbors on the West Coast of India and can shelter large ocean liners. The climate topography, geology and vegetation have played a prominent role in the development of the soils in the state. The soils in general are lateritic. The agricultural low lands, especially in the coastal track are alluvial flats formed through sedimentation along the principal rivers. In many cases, they are protected by dykes. Along the coastline there are sand dunes, too dry and barren to sustain any agricultural crop except cashew and coconut.

3.1.6 Most of the remaining land much of which classified as forests originates from Archaean rock formations which are very thick. In the highlands laterite covers goneiss and hypogenic rocks, from which iron ore is mined extensively. The soil is murum on the slopes. Many commercial tree species thrive in the well aerated soils at the foothills, where the murum is mixed with humus to form a loamy soil, and where the growth is often thick and high. In the valleys where the perennial natural water springs occur, arecanut plantations are seen.

3.1.7 The soils of this State can, therefore, be classified into three categories viz. (I) Laterite or Lateritic, (II) Alluvial and (III) Sandy. About 2, 75,900 hectares are of the first category and are highly acidic in nature, sandy loam to silt loam in texture and well drained. They are poor in lime, phosphorus and potash but fairly good in organic matter and nitrogen. About 24,950 hectares are alluvial, 17,600 hectares of which are subject to inundation of saline water and have to be protected by bunds to grow crops successfully. The nine kilometers coastal inland comprise a stretch of 38,000 hectares half of which has a high water table and could be exploited for irrigation and multiple cropping. These soils are also acidic, sandy to sandy loams, fairly rich in organic matter but very deficient in phosphate and potash.

3.2 Rainfall and temperature

In Goa, the climate is warm and humid. From June to September, heavy rainfall is received from the Southwest monsoons. There is much variation in temperature over the year. In the western half, where most of the land is below 100 meters in altitude, the rainfall average varies between 2600 and 4000 millimeters and the year round temperature varies from 18^0 C to 34^0 C. In the eastern half, where the hills and mountains rise to higher altitudes (max. 1,666 mts. above sea level), temperature variations are wider and the rainfall is considerably heavier. Among all the towns of the Goa State, the altitude of Mormugao town is the highest i.e. 62 mts above the sea level. Since the towns are situated very near to the coast, the maximum temperature seldom exceeds 34^0 C and the minimum seldom falls below 18^0 C.

3.3 Crop seasons

In the State of Goa, basically, the soil and climate conditions are favorable for the growth of a wide variety of tropical crops. Crops like paddy, ragi, pulses, beans, other millets and vegetables are grown during Kharif season (from May to August). The important crops which are grown during Rabi season (December to February) are paddy, pulses, beans and vegetables. During summer season some vegetables are grown in this State. Coconut and arecanut are important among these crops. The common fruit bearing trees are cashew, mango, jackfruit, breadfruit, papaya, lemon, banana and pineapple. Vegetables such as brinjal, ladies finger, radish, cucumber, pumpkin, drumsticks, different varieties of gourd etc. are widely grown. Sweet potatoes, onions and chilies are grown wherever irrigation facilities are available. Sugarcane is another crop introduced on a large scale due to establishment of sugar factory in the taluka of Dharbandora in South Goa District.

3.4 Population characteristics

As per the 2011 Population Census, the population of State of Goa was 14, 58,545. The Male population was recorded as 7,39,140 while that of Female was 7,19,405. The most populous town in the State is Mormugao with population of 94,393 and Shiroda is the highest populated village with population of 14,041. Jua has been registered as the least populated town having population of 4134. The Sex ratio in Goa stood at 973 during 2011 Population Census as against 961 in 2001 Population Census. Sex ratio of Child population of the age upto six years was 942 and above seven years was 977. The Literate population in Goa was recorded as 11, 65,487 (87.40%). Density Per Sq. Km. in Goa is found to be 394.

3.5 Main Characteristics of Agricultural Economy of the State

Agricultural sector plays a vital role in the distribution of regional income. A study of principal characteristics of this sector such as land utilisation, cropping pattern, extent of inputs and other related factors become very important from the point of view of proper planning at the micro level so that the share in the State income of particularly the small and the marginal farmers improves. A brief description of the agricultural economy of the State on the basis of available data is given below:

3.6 Contribution of Agriculture to State Income:

Agriculture proper contributes sizably to the regional income. The contribution of Agriculture Sector to the State Gross Domestic Estimates for the year 2010-11 as prepared by the Directorate of Planning, Statistics and Evaluation on the basis of standard methodology recommended by the Central Statistics Office of the Government of India; work out to Rs. 1,03,919.00 lakh at current prices which represent 3.09 percent of the total gross regional income. These estimates at constant prices (2004-05) works out to Rs. 62,990.00 lakh, i.e. 2.80 percent of the State Domestic Product for the year 2010-11.

3.6.1 The Net State Domestic Product at current prices and constant prices for Agriculture Sector estimated for the year 2010-11 was Rs. 88384.00 lakh and Rs. 53082.00 lakh respectively. The percentage to the total net income works out to 3.01 % and 2.75 % respectively at current and constant prices.

3.6.2 The total Gross Income at Current and Constant Prices during the last 2005-06 Agricultural Census (8th Census) was recorded at Rs. 84062.00 lakh and Rs. 82977.00 lakh respectively while net income during the said Census was Rs. 70485.00 lakh and Rs. 69966.00 lakh respectively.

3.6.3 The variation in income recorded during the 2010-11 Agricultural Census over the 2005-06 Census at Gross Income at Current and Constant Prices was (+) 23.62 % and (-) 24.09%, while at Net Income it was (+) 25.39% (Current) and (-) 24.13% (Constant) respectively.

CHAPTER IV

ANALYSIS OF THE CENSUS RESULT

4.1 The data collected under the main census, 2010-11, is tabulated in six tables. The data is grouped in 10 classes on the basis of size of operational holdings, the lowest size class being area operated below 0.5 hectares and the highest being 20 hectares and above. The grouping enables to study aspects such as land use, tenancy, cropping pattern and irrigation as related to the size of the holding. In this chapter, the analysis of the census result is presented and comparison with the results of the earlier census has also been attempted.

4.2 Operational Holdings and their Area.

According to the Agricultural Census, 2010-11 there were in all 78,020 operational holdings covering an area of 88,994 hectares. There were 52,821 operational holdings covering an area of 60,742 hectares during the Agricultural Census 2005-06. Thus, there was an increase of 48 percent in the number of operational holdings and increase of 47 percent in the area during the 2010-11 Census, as compared to the Census of 2005-06.

4.3 Size Distribution of Operational Holdings

4.3.1 In the table 4.1 below, the distribution of number of operational holdings and their corresponding area during the year 2010-11 is presented according to different size classes of operational holdings.

TABLE 4.1
DISTRIBUTION OF OPERATIONAL HOLDINGS AND AREA ACCORDING TO THE SIZE CLASSES DURING 2010-2011

Sl. No.	Size Class	No. of operational holdings			Area. of operational holdings		
		Total	Percentage	Cumulative Percentage	Total	Percentage	Cumulative Percentage
1	2	3	4	5	6	7	8
1	Below 0.5	42414	54.36	54.36	14107	15.85	15.85
2	0.5-1.0	17486	22.41	76.77	13996	15.73	31.58
3	1.0-2.0	9817	12.58	89.35	17591	19.77	51.35
4	2.0-3.0	4397	5.64	94.99	12090	13.58	64.93
5	3.0-4.0	1310	1.68	96.67	4680	5.26	70.19
6	4.0-5.0	902	1.16	97.83	4207	4.73	74.92
7	5.0-7.5	775	0.99	98.82	5241	5.89	80.81
8	7.5-10.0	333	0.43	99.25	2930	3.29	84.10
9	10.0-20.0	399	0.51	99.76	6138	6.90	91.00
10	20.0 & above	187	0.24	100.00	8014	9.00	100.00
Total		**78020**	**100.00**		**88994**	**100.00**	

4.3.2 The above table reveals that the size distribution of the number of operational holdings is positively skewed with the highest frequency (54.36 percent) in the lowest size class (0.5 ha.). The size classes upto 2.0 ha. and below accounted for almost (89.35 percent) of the total number of holdings. The holdings of the size of above 10.0 ha. formed only about 0.75 percentage of the total number of holdings. The area of operational holdings of size 2.0 ha. and below formed 51.35 percent of the total operated area and that of operational holdings above 10.0 ha. formed 15.90 percent of the total operated area.

4.3.3 The distribution of number of operational holdings and their corresponding area during 2010-11 Census presented in the table 4.2 below by major size classes and groups viz. marginal farmers (below 1.0 ha.), small farmers (1.0 – 2.0 ha.), semi medium farmers (2.0 – 4.0 ha.), medium farmers (4.0 – 10.0 ha.) and large farmers (above 10.0 ha.). An attempt is also made to compare the data, with the corresponding data of 2005-06 census.

TABLE 4.2

DISTRIBUTION OF NUMBER AND AREA OF OPERATIONAL HOLDINGS BY FIVE MAJOR CLASSES OF HOLDINGS IN 2005-06 AND 2010-11

SN	Major Size Classes	Size Group (in Ha.)	Number of Operational Holdings				Area (In Ha.) of Operational Holdings			
			Numbers		Percentages		Hectares		Percentages	
			2005-06	2010-11	2005-06	2010-11	2005-06	2010-11	2005-06	2010-11
1	2	3	5	6	7	8	9	10	11	12
1	Marginal	Below 1.0	42745	59900	80.92	76.77	12422	28103	20.45	31.58
2	Small	1.0-2.0	5788	9817	10.96	12.58	7183	17591	11.83	19.77
Sub-Total		Below 2.0	48533	69717	91.88	89.35	19605	45694	32.28	51.35
3	Semi-Medium	2.0-4.0	2681	5707	5.08	7.32	6725	16770	11.07	18.84
Sub-Total		Below 4.0	51214	75424	96.96	96.67	26330	62464	43.35	70.19
4	Medium	4.0-10.0	1195	2010	2.26	2.58	6813	12378	11.22	13.91
5	Large	Above 10.0	412	586	0.78	0.75	27599	14152	45.43	15.90
All Categories			**52821**	**78020**	**100.00**	**100.00**	**60742**	**88994**	**100.00**	**100.00**

4.3.4 The distribution of number and area of the operational holdings according to the five classes indicated above shows a preponderance of marginal farmers. The holdings of less than one hectare accounted for more than three fourths (76.77 percent) of the total holdings in the State, whereas the area covered by these operational holdings formed only 31.58 percent during 2010-11of the total operated area. If small farmers are also taken into account along with the marginal farmers then the percentage of these holdings to the total number of holdings was observed to be 89.35 percent and the area operated by these

two categories of holdings (upto 2.0 hectares) worked out to 51.35 percent. If the holdings upto 4.0 hectares are considered, then it covered 96.67 percent of the total holdings and 70.19 percent of the total operated area. In the upper strata the holdings of more than 10 hectares were only 0.75 percent but their area coverage was almost 15.90 percent. During the 2005-06 census 1528 holdings or 2.89 percent were joint holdings but according to the present census 2279 holdings or 2.92 percent fall under this category.

4.3.5 The average size of the operational holding by major size classes of holdings in the year 2005-06 & 2010-11 is presented in the table 4.3 below.

TABLE 4.3

AVERAGE SIZE OF OPERATIONAL HOLDINGS BY FIVE MAJOR CLASSES IN 2005-06 AND 2010-11

Sr. No.	Size Class	2005-2006			2010-11		
		No. of Holdings	Area (in Ha.)	Average size	No. of Holdings	Area (in Ha.)	Average size
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**
1	Below 1.0	42745	12422	0.29	59900	28103	0.47
2	1.0-2.0	5788	7183	1.24	9817	17591	1.79
3	2.0-4.0	2681	6725	2.51	5707	16770	2.94
4	4.0-10.0	1195	6813	5.70	2010	12378	6.16
5	10.0 & above	412	27599	66.99	586	14152	24.15
	Total	**52821**	**60742**	**1.15**	**78020**	**88994**	**1.14**

4.3.6 The average area of holding by categories of holdings, that is marginal, small, semi-medium, and medium holdings has increased during 2010-11 as compared to 2005-06, but the average holding area for upper size class 10.0 ha and above has decreased from 66.99 to 24.15. The average size of holding for the State which was 1.15 ha. in 2005-06 is 1.14 ha. in 2010-11.

4.4 Individual, Joint and Institutional Holdings

4.4.1 Out of the total 78020 holdings in the State 75616 (96.92 percent) were individual holdings with an area coverage of 81347 hectares (91.41 percent). The joint holdings were 2279 i.e. 2.92 percent covering an area of only 5596 hectares (6.29 percent). There were 125 institutional holdings i.e. 0.16 percent with area coverage of 2051 hectares i.e. 2.30 percent.

TABLE 4.4

DISTRIBUTION OF NUMBER AND AREA OF OPERATIONAL HOLDINGS ACCORDING TO INDIVIDUAL, JOINT AND INSTITUTIONAL HOLDINGS IN 2010-2011

Sl. No.	Category	Number	Percentage	Area	Percentage
1	**2**	**3**	**4**	**5**	**6**
1	Individual	75616	96.92	81347	91.41
2	Joint	2279	2.92	5596	6.29
3	Institutional	125	0.16	2051	2.30
Total		**78020**	**100.00**	**88994**	**100.00**

The district-wise number of holdings and the area under these holdings according to the 2010-11 Census are given in the following table.

TABLE 4.5

DISTRICTWISE DISTRIBUTION OF NUMBER AND AREA OF OPERATIONAL HOLDINGS IN 2010-2011

Districts	No. of holdings	Percentage to total no. of holdings	Area operated in Hectares	Percentages to area operated	Average size of the operational holdings (ha)
1	**2**	**3**	**4**	**5**	**6**
North Goa	45891	58.82	51375	57.73	1.12
South Goa	32129	41.18	37619	42.27	1.17
Total	**78020**	**100.00**	**88994**	**100.00**	**1.14**

4.4.2 It is seen from the above table that North Goa District recorded 45891 (58.82 percent) holdings accounting for 57.73 percent of the total operated area in the State. There were 32129 (41.18 percent) holdings in the South Goa District covering 42.27 percent of the total operated area in the State. The average size of operational holdings had no much difference.

4.4.3 District wise and Taluka wise distribution of operational holdings by size classes is presented in the table 4.6 (A) and 4.6 (B) below.

TABLE 4.6(A)
DISTRICTWISE DISTRIBUTION OF OPERATIONAL HOLDINGS BY CLASSES IN 2010-2011

State/ Districts	No. of holdings & area operated	Less than 1.0 ha. (Marginal)	1.0 to 2.0 ha. (Small)	2.0 to 4.0 ha. (Semi Medium)	4.0 to 10.0 ha. (Medium)	10.0 ha. & above (Large)	Total
1	2	3	4	5	6	7	8
Goa State	Number	59900	9817	5707	2010	586	78020
		(76.78)	(12.58)	(7.31)	(2.58)	(0.75)	(100.00)
	Area	28103	17591	16770	12378	14152	88994
		(31.58)	(19.77)	(18.84)	(13.91)	(15.90)	(100.00)
North Goa	Number	36520	4994	2901	1102	374	45891
		(79.58)	(10.88)	(6.32)	(2.40)	(0.82)	(100.00)
	Area	17326	8865	8554	6951	9679	51375
		(33.72)	(17.26)	(16.65)	(13.53)	(18.84)	(100.00)
South Goa	Number	23380	4823	2806	908	212	32129
		(72.77)	(15.01)	(8.73)	(2.83)	(0.66)	(100.00)
	Area	10777	8726	8216	5427	4473	37619
		(28.65)	(23.19)	(21.84)	(14.43)	(11.89)	(100.00)

Note: Figures in brackets indicate percentages

TABLE 4.6 (B)

TALUKAWISE DISTRIBUTION OF OPERATIONAL HOLDINGS BY CLASSES IN 2010-2011

Taluka/ District/ State	No. of Operation -al holding	Less than 1.0 ha (Marginal)	1.0 to 2.0 ha (Small)	2.0 to 4.0 ha (Semi Medium)	4.0 to 10.0 ha. (Medium)	10.0 ha. & above (Large)	Total
1	2	3	4	5	6	7	8
Tiswadi	Number	2544	300	134	81	14	3073
		(82.78)	(9.76)	(4.36)	(2.64)	(0.46)	(100.00)
	Area	1402	571	392	518	358	3241
		(43.26)	(17.62)	(12.10)	(15.98)	(11.04)	(100.00)
Bardez	Number	6739	937	297	63	21	8057
		(83.64)	(11.63)	(3.69)	(0.78)	(0.26)	(100.00)
	Area	4060	1675	869	429	365	7398
		(54.88)	(22.64)	(11.75)	(5.80)	(4.93)	(100.00)
Pernem	Number	7939	952	415	136	16	9458
		(83.94)	(10.06)	(4.39)	(1.44)	(0.17)	(100.00)
	Area	3838	1667	1198	862	279	7844
		(48.93)	(21.25)	(15.27)	(10.99)	(3.56)	(100.00)
Bicholim	Number	6668	638	473	229	99	8107
		(82.25)	(7.87)	(5.84)	(2.82)	(1.22)	(100.00)
	Area	2453	1135	1370	1423	2416	8797
		(27.89)	(12.90)	(15.57)	(16.18)	(27.46)	(100.00)
Sattari	Number	5021	1467	1092	381	141	8102
		(61.97)	(18.11)	(13.48)	(4.70)	(1.74)	(100.00)
	Area	2798	2569	3301	2441	4133	15242
		(18.36)	(16.85)	(21.66)	(16.01)	(27.12)	(100.00)
Ponda	Number	7609	700	490	212	83	9094
		(83.67)	(7.70)	(5.39)	(2.33)	(0.91)	(100.00)
	Area	2775	1249	1424	1278	2127	8853
		(31.34)	(14.11)	(16.08)	(14.44)	(24.03)	(100.00)
North Goa (District)	Number	**36520**	**4994**	**2901**	**1102**	**374**	**45891**
		(79.58)	**(10.88)**	**(6.32)**	**(2.40)**	**(0.82)**	**(100.00)**
	Area	**17326**	**8865**	**8554**	**6951**	**9679**	**51375**
		(33.72)	**(17.26)**	**(16.65)**	**(13.53)**	**(18.84)**	**(100.00)**

Note: Figures in brackets indicate percentages

Taluka/ District/ State	No. of Operatio-nal holdings	Less than 1.0 ha (Marginal)	1.0 to 2.0 ha (Small)	2.0 to 4.0 ha (Semi Medium)	4.0 to 10.0 ha. (Medium)	10.0 ha. & above (Large)	Total
Sanguem	Number	4439	1275	505	146	43	6408
		(69.27)	(19.90)	(7.88)	(2.28)	(0.67)	**(100.00)**
	Area	1911	2257	1493	795	927	7383
		(25.88)	(30.57)	(20.22)	(10.77)	(12.56)	**(100.00)**
Salcete	Number	8960	729	281	60	41	10071
		(88.97)	(7.24)	(2.79)	(0.59)	(0.41)	**(100.00)**
	Area	3424	1331	814	323	782	6674
		(51.30)	(19.94)	(12.20)	(4.84)	(11.72)	**(100.00)**
Quepem	Number	4951	1104	561	196	44	6856
		(72.22)	(16.10)	(8.18)	(2.86)	(0.64)	**(100.00)**
	Area	2208	1927	1630	1162	941	7868
		(28.06)	(24.49)	(20.72)	(14.77)	(11.96)	**(100.00)**
Mormugao	Number	848	9	14	24	2	897
		(94.54)	(1.00)	(1.56)	(2.68)	(0.22)	**(100.00)**
	Area	542	16	44	120	31	753
		(71.99)	(2.12)	(5.84)	(15.93)	(4.12)	**(100.00)**
Canacona	Number	3124	1270	1203	418	42	6057
		(51.58)	(20.97)	(19.86)	(6.90)	(0.69)	**(100.00)**
	Area	2012	2432	3518	2618	654	11234
		(17.91)	(21.65)	(31.32)	(23.30)	(5.82)	**(100.00)**
Dharbandora	Number	1058	436	242	64	40	1840
		(57.50)	(23.70)	(13.15)	(3.48)	(2.17)	**(100.00)**
	Area	680	763	717	409	1138	3707
		(18.34)	(20.58)	(19.34)	(11.03)	(30.70)	**(100.00)**
South Goa (District)	Number	**23380**	**4823**	**2806**	**908**	**212**	**32129**
		(72.77)	**(15.01)**	**(8.73)**	**(2.83)**	**(0.66)**	**(100.00)**
	Area	**10777**	**8726**	**8216**	**5427**	**4473**	**37619**
		(28.65)	**(23.19)**	**(21.84)**	**(14.43)**	**(11.89)**	**(100.00)**
Goa State (Total)	Number	**59900**	**9817**	**5707**	**2010**	**586**	**78020**
		(76.78)	**(12.58)**	**(7.31)**	**(2.58)**	**(0.75)**	**(100.00)**
	Area	**28103**	**17591**	**16770**	**12378**	**14152**	**88994**
		(31.58)	**(19.77)**	**(18.84)**	**(13.91)**	**(15.90)**	**(100.00)**

Note: Figures in brackets indicate percentages

TABLE 4.6 (C)
TALUKAWISE DISTRIBUTION OF NUMBER & AREA OF INDIVIDUAL, JOINT & INSTITUTIONAL OPERATIONAL HOLDINGS BY MAJOR CLASSES DURING 2010-2011

Taluka/ District/ State	Number and area	Less than 1.0 ha. (Marginal)			1.0 to 2.0 ha. (Small)			2.0 to 4.0 ha. (Semi-Medium)			4.0 to 10.0 ha. (Medium)			10.0 ha. & above (Large)			Total (All Classes)		
		Indiv	Joint	Inst.	Indiv	Joint	Inst.	Indiv	Joint	Inst.	Indiv	Joint	Inst.	Indiv	Joint	Inst.	Indiv	Joint	Inst.
Tiswadi	No.	2238	302	4	251	45	4	100	30	4	53	17	11	7	7	0	2649	401	23
	Area	1265	134	3	485	80	6	293	87	13	353	98	67	173	185	0	2569	584	89
Bardez	No.	6511	226	2	829	108	0	258	39	0	56	7	0	19	2	0	7673	382	2
	Area	3919	140	1	1486	190	0	757	112	0	387	41	0	337	28	0	6886	511	1
Pernem	No.	7841	88	10	898	54	0	403	12	0	124	12	0	16	0	0	9282	166	10
	Area	3779	55	4	1571	95	0	1165	33	0	783	79	0	279	0	0	7577	263	4
Bicholim	No.	6481	180	7	585	53	0	428	41	4	203	26	0	74	23	2	7771	323	13
	Area	2343	106	4	1041	94	0	1243	118	10	1242	181	0	1829	561	26	7698	1060	40
Sattari	No.	4845	176	0	1402	65	0	994	98	0	348	33	0	95	36	10	7684	408	10
	Area	2697	100	0	2456	113	0	3015	285	0	2224	217	0	2074	787	1272	12466	1503	1272
Ponda	No.	7413	194	2	669	25	6	482	5	3	208	4	0	76	7	0	8848	235	11
	Area	2690	84	1	1195	44	10	1398	18	8	1248	29	0	1465	663	0	7996	838	19
District North Goa	No.	35329	1166	25	4634	350	10	2665	225	11	992	99	11	287	75	12	43907	1915	69
	Area	16694	620	12	8233	616	16	7870	653	31	6239	645	67	6156	2224	1299	45192	4757	1425
Sanguem	No.	4439	0	0	1272	0	3	502	0	3	141	3	2	38	0	5	6392	3	13
	Area	1911	0	0	2253	0	5	1484	0	8	763	13	20	713	0	214	7124	13	247
Salcete	No.	8889	65	6	716	11	2	275	3	3	53	7	0	28	11	2	9961	97	13
	Area	3383	37	3	1308	19	4	797	8	8	282	41	0	577	176	29	6348	281	44
Quepem	No.	4890	59	2	1091	13	0	548	13	0	187	5	4	36	6	2	6752	96	8
	Area	2171	36	1	1907	20	0	1593	38	0	1102	26	33	776	96	69	7549	216	102
Mormugao	No.	842	0	6	9	0	0	14	0	0	11	11	2	2	0	0	878	11	8
	Area	538	0	4	16	0	0	44	0	0	55	52	13	31	0	0	684	52	17
Canacona	No.	3086	34	4	1181	89	0	1200	0	3	405	10	3	42	0	0	5914	133	10
	Area	2003	10	1	2275	155	0	3509	0	10	2536	70	11	653	0	0	10976	235	22
Dharbandora	No.	1043	15	0	433	3	0	238	4	0	62	2	0	36	0	4	1812	24	4
	Area	673	7	0	759	4	0	707	10	0	389	20	0	946	0	192	3474	41	192
District South Goa	No.	23189	173	18	4702	116	5	2777	20	9	859	38	11	182	17	13	31709	364	56
	Area	10679	90	8	8519	198	9	8134	56	26	5127	221	79	3696	273	504	36155	838	625
Goa State	No.	58518	1339	43	9336	466	15	5442	245	20	1851	137	22	469	92	25	75616	2279	125
	Area	27373	710	20	16752	814	25	16004	709	57	11366	866	146	9852	2497	1803	81347	5596	2051

*Totals may not tally exactly due to rounding off

The percentages of number and area of marginal holders was higher in North Goa District 60.97% and 61.65% when compared with South Goa District 39.03% and 38.35% . The number and area percentages of small holders were placed as 50.87% and 50.40% and 49.13% and 49.60% in North Goa & South Goa District respectively. These are the two major classes recognized during the Agricultural Census 2010-11.

4.4.4 The following table gives the average size of holdings in different districts during 2005-06 & 2010-11.

TABLE 4.7

DISTRICTWISE AVERAGE SIZE OF HOLDINGS IN 2005-06 & 2010-11

Name of the District	Agricultural Census 2005-06			Agricultural Census 2010-11		
	Operational holdings (No.)	Area (Ha)	Average size (Ha)	Operational holdings (No.)	Area (Ha)	Average size (Ha)
1	2	3	4	5	6	7
North Goa	28748	24503	0.85	45891	51375	1.12
South Goa	24073	36239	1.51	32129	37619	1.17
Goa State	**52821**	**60742**	**1.15**	**78020**	**88994**	**1.14**

4.4.5 The average size of the holding is found to be in close vicinity during the current Census (1.14 ha.) as compared to 2005-06 (1.15 ha.). The inter-district comparison was slightly different, as in South Goa District, it was 1.51 ha. in 2005-06 which is decreased to 1.17 ha. in 2010-11. While in North Goa District it increased to 1.12 ha. in 2010-11 from 0.85 ha. in 2005-06. The number and area of holdings were found increased to 60 % and 110% respectively in North District during 2010-11 as compared to 2005-06. The increase in South Goa was much slower.

4.5 Tenancy of Operational Holdings

4.5.1 Distribution of number of operational holdings according to tenancy status by major size classes is given in the following table 4.8

TABLE 4.8

DISTRIBUTION OF NUMBER OF OPERATIONAL HOLDINGS ACCORDING TO TENANCY STATUS BY MAJOR SIZE CLASSES 2010-11

Sl. No.	Major size class (in ha.)	Total No. of holdings	Wholly owned and self operated	Wholly leased in	Wholly otherwise operated	Partly owned, Partly leased in and partly otherwise operated
1	2	3	4	5	6	7
1	Marginal	59900	51692	4893	2134	1181
	(below 1.0)	(76.78)	(66.26)	(6.27)	(2.74)	(1.51)
2	Small	9817	8016	560	292	949
	(1.0 to 2.0)	(12.58)	(10.27)	(0.72)	(0.37)	(1.22)
3	Semi-Medium	5707	4211	383	97	1016
	(2.0 to 4.0)	(7.31)	(5.40)	(0.49)	(0.12)	(1.30)
4	Medium	2010	1560	77	33	340
	(4.0 to 10.0)	(2.58)	(2.00)	(0.10)	(0.04)	(0.44)
5	Large	586	558	5	0	23
	(10.0 above)	(0.75)	(0.71)	(0.01)	(0.00)	(0.03)
Total		**78020**	**66037**	**5918**	**2556**	**3509**
		(100.00)	**(84.64)**	**(7.58)**	**(3.28)**	**(4.50)**

Note: Figures in brackets indicate percentages.

4.5.2 The following table gives the distribution of area operated according to tenancy by size classes.

TABLE 4.9

DISTRIBUTION OF AREA OPERATED ACCORDING TO TENANCY BY MAJOR SIZE CLASSES 2010-11

(in Ha)

Sl. No.	Major Size Classes	Total Operated Area	Wholly Owned And Self Operated	Wholly Leased In	Wholly Otherwise Operated	Partly Owned, Partly Leased In And Partly Otherwise Operated			
						Owned	Leased In	Other-wise Operated	Total
1	2	3	4	5	6	7	8	9	10
		28103	23706	2608	1017	331	253	188	772
1	Marginal	(31.58)	(32.23)	(49.14)	(50.65)	(0.37)	(0.28)	(0.21)	(0.87)
		17591	14165	1037	546	695	458	691	1843
2	Small	(19.77)	(19.26)	(19.54)	(27.22)	(0.78)	(0.51)	(0.78)	(2.07)
	Semi	16770	12307	1116	292	1190	915	950	3055
3	Medium	(18.84)	(16.73)	(21.03)	(14.56)	(1.34)	(1.03)	(1.07)	(3.43)
		12378	9711	457	152	1200	411	446	2058
4	Medium	(13.91)	(13.20)	(8.61)	(7.58)	(1.35)	(0.46)	(0.50)	(2.31)
		14152	13663	89	0	288	49	63	400
5	Large	(15.90)	(18.58)	(1.68)	(0.00)	(0.32)	(0.05)	(0.07)	(0.45)
		88994	73552	5307	2007	3704	2086	2338	8128
All Categories		(100.00)	(82.65)	(5.96)	(2.25)	(4.16)	(2.34)	(2.63)	(9.14)

Note: Figures in brackets indicates percentages.
Total may not tally exactly due to rounding off.

4.5.3 It can be observed that the bulk of holdings and area operated (84.64 percent and 82.65 percent respectively) was accounted by wholly owned and self operated category. Next to this 7.58 percent of the operational holdings were leased in accounting for 5.96 percent of total operated area.

4.5.4 About 3.28 percent of the holdings belonged to the category 'wholly otherwise operated' and these accounted for 2.25 percent of the total area operated. The category partly owned, partly leased in and partly otherwise operated which accounted for 4.50 percent of the holdings had a share of 9.14 percent in the total area operated under this category. The number/percentage of holdings in Table No. 4.8 and operated area/percentages in Table No. 4.9 may not tally with the totals of selected major size classes presented due to rounding.

4.5.5 District-wise percentage share by tenancy status is presented in the following table 4.10

TABLE 4.10

DISTRICT-WISE PERCENTAGE SHARE BY SELECTED TENANCY STATUS IN GOA 2010-11

Sl. No.	Name of the District	Wholly Owned and Self Operated		Wholly Leased In		Wholly Otherwise Operated		Partly Owned, Partly Leased In and Partly Otherwise Operated	
		No.	Area	No.	Area	No.	Area	No.	Area
1	2	3	4	5	6	7	8	9	10
		38977	44571	4174	3821	1773	1258	967	1727
1	North Goa	(59.02)	(60.60)	(70.53)	(72.00)	(69.37)	(62.68)	(27.56)	(21.25)
		27060	28981	1744	1486	783	749	2542	6401
2	South Goa	(40.98)	(39.40)	(29.47)	(28.00)	(30.63)	(37.32)	(72.44)	(76.75)
		66037	**73552**	**5918**	**5307**	**2556**	**2007**	**3509**	**8128**
Goa State		**(100.00)**	**(100.00)**	**(100.00)**	**(100.00)**	**(100.00)**	**(100.00)**	**(100.00)**	**(100.00)**

Note: Figures in brackets are in percentages.
Total may not tally exactly due to rounding off.

4.5.6 Wholly owned and self operated holdings in North Goa District were 59.02 percent of the total holdings and 60.60 percent of the operated area, while in South Goa District there were 40.98 percent holdings with 39.40 percent share in the total operated area.

4.6 LAND UTILISATION

4.6.1 According to the Agricultural Census 2010-11 total operated area of all the holdings was 88994 hectares. Out of this area 87214 hectares or 98.00 percent was net sown area and 305 hectares or 0.34 percent was current fallow. Thus the net cultivated area formed 98.34 percent of the total operational area in the State. The share of the cultivable waste land was 353 ha. or 0.40 percent. The proportionate share of the land not available for cultivation was 509 ha. or 0.57 percent. It would be quite interesting to study the various relationships such as holdings size and land utilisation size of the holdings and intensity of cropping etc. Table 4.11 below gives the area under different uses by size classes.

TABLE 4.11

DISTRIBUTION OF AREA UNDER LAND BY MAJOR SIZE CLASSES IN 2010-11

SL. No.	Size Classes (ha)	Net area sown	Current fallow	Net cultivated area	Other unculti-vated land excluding fallow	Fallow land other than current fallow	Cultiv-able waste land	Total unculti-vated land	Land not availa-ble for cultiva-tion
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
1	Below 1.0	27853	65	27918	46	26	63	135	49
	(Marginal)	(31.94)	(21.31)	(31.90)	(13.77)	(9.32)	(17.85)	(13.98)	(9.65)
2	1.0-2.0	17332	41	17373	86	24	64	174	44
	(Small)	(19.87)	(13.44)	(19.85)	(25.75)	(8.60)	(18.13)	(18.01)	(8.66)
3	2.0 – 4.0	16471	81	16552	87	39	62	188	31
	(Semi-Medium)	(18.88)	(26.56)	(18.91)	(26.05)	(13.98)	(17.56)	(19.46)	(6.10)
4	4.0 – 10.0	12044	78	12122	65	35	98	197	59
	(Medium)	(13.81)	(25.57)	(13.85)	(19.46)	(12.54)	(27.76)	(20.39)	(11.42)
5	10.0 & above	13514	40	13554	51	155	66	272	326
	(Large)	(15.50)	(13.12)	(15.49)	(14.97)	(55.56)	(18.70)	(28.06)	(64.17)
Total		**87214**	**305**	**87519**	**334**	**279**	**353**	**966**	**509**
		(100.00)	**(100.00)**	**(100.00)**	**(100.00)**	**(100.00)**	**(100.00)**	**(100.00)**	**(100.00)**

Note: Figures in brackets indicate percentages.

4.6.2 A study of land utilization by size classes of holdings reveals that 31.94 percent of the net area sown was operated by marginal farmers, while small farmers had 19.87 percent of area under operation. 32.69 percent area was operated by semi medium and medium size holders. Large size classes cultivated only 15.50 percent of the total net area sown. The combined share of area under current fallow, total uncultivated area and land not available for cultivation was 1780 ha or 2.00 percent of the total area of the holdings.

4.6.3 Percentage distribution of area by size classes according to land utilization is given in the table 4.12 below.

TABLE 4.12

PERCENTAGE DISTRIBUTION OF AREA BY MAJOR SIZE CLASSES ACCORDING TO LAND UTILISATION IN 2005-06 AND 2010-11

Sl. No.	Size class (ha.)	Net Cultivated Area		Other Area	
		2005-06	2010-11	2005-06	2010-11
1	2	3	4	5	6
1	Below 1.0 (Marginal)	99.85	99.34	0.15	0.66
2	1.0-2.0 (Small)	99.62	98.76	0.38	1.24
3	2.0 – 4.0 (Semi-Medium)	99.27	98.70	0.73	1.30
4	4.0 – 10.0 (Medium)	98.93	97.93	1.07	2.07
5	10.0 & above (Large)	94.32	95.78	5.68	4.22
All sizes		**97.14**	**98.34**	**2.86**	**1.66**

4.6.4 A study of the table 4.12 above clearly shows that there were only minor differences between the results of two censuses as far as percentage distribution of area according to different size classes is concerned. The percentage of net cultivated area during 2010-11 was slightly higher than it was in the last census.

4.7 NET AREA SOWN

4.7.1 District wise distribution of area according to the Land utilization is presented in the following table 4.13

TABLE 4.13

DISTRICTWISE DISTRIBUTION OF AREA ACCORDING TO THE LAND UTILISATION IN 2010-11

Sl. No.	Name of the District	Land Utilization							
		Net area sown	Current fallow	Net cultiv-ated area	Other uncultiv-ated land excluding fallow	Fallow land other than current fallow	Cultiv able waste land	Total uncult-ivated land	Land not avail-able for cultiv-ation
1	2	3	4	5	6	7	8	9	10
1	North Goa	50179	191	50370	160	250	251	661	344
		(56.38)	(0.21)	(56.60)	(0.18)	(0.28)	(0.28)	(0.74)	(0.39)
2	South Goa	37035	114	37149	174	29	102	305	165
		(41.62)	(0.13)	(41.74)	(0.20)	(0.03)	(0.11)	(0.34)	(0.18)
Goa State		**87214**	**305**	**87519**	**334**	**279**	**353**	**966**	**509**
		(98.00)	**(0.34)**	**(98.34)**	**(0.38)**	**(0.31)**	**(0.40)**	**(1.09)**	**(0.57)**

Note: Figures in brackets indicate percentages.

4.7.2 In Goa State the net area sown was 87214 hectares during 2010-11 Agriculture Census, which formed 98.00 percent of the total operated area (88994 ha). Its distribution in North Goa and South Goa was 50179 hectares & 37035 hectares respectively.

4.8 CURRENT FALLOW:

Current fallow represents land which remained fallow during the particular agricultural year in discussion. Sometimes the cultivators allow the land to remain fallow during the current year to revive the fertility of their lands while with others, it may remain fallow due to lack of rainfall, seed problem etc. During the agricultural census year 2010-11, the area of current fallow constituted 0.34 percent of the total operated area. From the table 4.13 it is clear that the proportion of land under current fallow was very low in both the districts, viz North Goa and South Goa, which was recorded at 0.21 percent and 0.13 percent respectively.

4.9 NET CULTIVATED AREA

The net area sown and current fallow taken together represent the net cultivated area. During 2010-11 Agriculture Census, the area covered under this category was 87519 hectares of land and formed 98.34 percent of the total area of holdings. Out of 87519 hectares of net area cultivated, about 57.55 percent was accounted for by the North Goa District and 42.45 percent by South Goa District.

4.10 TOTAL UNCULTIVATED LAND

The total uncultivated land, which comprises (i) other uncultivated land excluding fallow land, (ii) fallow other than current fallow and (iii) cultivable wasteland, formed 1.09 percent of the total area. From the land utilization point of view, this category, also occupied an important place, as the vegetation that grew naturally in this area provided fodder to the live stock, because specially grass is normally not grown in this State as a crop. The permanent pastures in Goa State provide grass to Live Stock. The percentage distribution of uncultivated land in North and South districts was 68 % and 32 % respectively of the uncultivated land and 0.74 % and 0.34 % of the total land holding in the State.

4.11 AREA NOT AVAILABLE FOR CULTIVATION

It covered only 0.57 percent of the total operated area in the State. The percentage distribution of such area in North Goa and South Goa District was 0.39 and 0.18 respectively of the total land holdings in the State.

4.12 INTENSITY OF CROPPING

The agro-climatic conditions and the pattern of land holding being peculiar in the State, extensive cultivation is neither possible nor practicable. The intensity of cropping for the State worked out to 101.07 percent, also both the districts had about the similar percentages. District wise details of the intensity of cropping in the State are given in the table 4.14.

TABLE 4.14

DISTRICTWISE DISTRIBUTION OF INTENSITY OF CROPPING (Percentage)

Sl. No.	Name of the District	Net area sown (ha)	Gross cropped area (ha)	Intensity of cropping col. 4 over 3* 100
1	*2*	*3*	*4*	*5*
1	North Goa	50179	51048	101.73
2	South Goa	37035	37096	100.16
Goa State		**87214**	**88144**	**101.07**

4.13 HOLDING BY IRRIGATION STATUS AND AREA IRRIGATED

4.13.1 The current Census has revealed that excluding the purely current fallow holdings, there were 40007 holdings receiving irrigation while 37958 holdings were wholly unirrigated. Out of the total irrigated holdings, 30468 holdings were wholly irrigated and 9539 holdings were partly irrigated.

4.13.2 The following table 4.15 furnishes the distribution of number and area of holdings by irrigation status in 2010-11.

TABLE 4.15

DISTRIBUTION OF NUMBER AND AREA OF HOLDINGS BY IRRIGATION STATUS 2010-11

Sl. No.	Irrigation status of the holding	Number of holdings	Area of holdings	
			Net area sown	Net irrigated area
1	*2*	*3*	*4*	*5*
1	Wholly Irrigated	30468	27097	27097
2	Partly Irrigated	9539	20366	8473
	Sub-Total (1+2)	40007	47463	35570
3	Wholly Un-Irrigated	37958	39751	0
Irrigated and Un-irrigated holdings		**77965**	**87214**	**35570**

4.13.3 The net area sown in these holdings was 87214 hectares. Of this 47463 hectares were irrigated consisting of 27097 hectares in the wholly irrigated holdings and another 20366 hectares in the partly irrigated holdings.

4.13.4 Net irrigated area and net area sown by size group of holdings 2010-11 is furnished in the following table 4.16

TABLE 4.16

NET IRRIGATED AREA AND NET AREA SOWN BY MAJOR SIZE CLASSES OF HOLDINGS IN 2010-11

Sl. No.	Category of holdings	Area of holding		Percentage of net irrigated area to net area sown
		Net area sown	Net area irrigated	
1	2	3	4	5
1	Below 1.0 (Marginal)	27853	11443	41.08
2	1.0 - 2.0 (Small)	17332	7244	41.80
3	2.0 - 4.0 (Semi Medium)	16471	5698	34.59
4	4.0 – 10.0 (Medium)	12044	4704	39.06
5	10.0 & above (Large)	13514	6481	47.96
All categories		**87214**	**35570**	**40.78**

4.13.5 It is seen that the percentage of net irrigated area to net area sown, which is an important indicator of the availability of irrigation for crop production, was maximum of 47.96 percent in case of large holdings and minimum of 34.59 percent incase of semi medium holdings. Average for all types categories was 40.78 percent.

4.13.6 Percentage distribution of the number and net irrigated area of holdings by irrigation status and size classes is given in the following table 4.17

TABLE 4.17

PERCENTAGE DISTRIBUTION OF NUMBER AND NET IRRIGATED AREA OF HOLDINGS BY IRRIGATION STATUS IN MAJOR SIZE CLASSES IN 2010-11

Sl. No.	Size Class	Wholly Irrigated		Partly Irrigated Total			Wholly Unirrigated	
		No. of holdings	Net area sown	No. of holdings	Net Area Irrigated	Net area sown	No. of holdings	Net area sown
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
1	Below 1.0 (Marginal)	24791	9940	5567	1504	3471	29490	14442
		(81.37)	(36.68)	(58.36)	(17.75)	(17.04)	(77.69)	(36.33)
2	1.0-2.0 (Small)	3480	5822	1823	1422	3286	4511	8224
		(11.42)	(21.49)	(19.11)	(16.78)	(16.14)	(11.88)	(20.69)
3	2.0 – 4.0 (Semi-Medium)	1436	4027	1359	1671	4045	2912	8399
		(4.71)	(14.86)	(14.25)	(19.72)	(19.86)	(7.67)	(21.13)
4	4.0 – 10.0 (Medium)	563	3312	563	1391	3367	884	5364
		(1.85)	(12.22)	(5.90)	(16.42)	(16.53)	(2.33)	(13.49)
5	10.0 & above (Large)	198	3996	227	2485	6197	161	3322
		(0.65)	(14.75)	(2.38)	(29.33)	(30.43)	(0.43)	(8.36)
Total		**30468**	**27097**	**9539**	**8473**	**20366**	**37958**	**39751**
		(100.00)	**(100.00)**	**(100.00)**	**(100.00)**	**(100.00)**	**(100.00)**	**(100.00)**

Note: Figures in brackets indicate percentages.

4.13.7 The above table reveals that the maximum 81.37 percent of the total number of wholly irrigated holdings belongs to the lowest size class of marginal holdings and only 0.65 percent of the holdings were large holdings. In terms of area, the marginal holdings accounted for maximum 36.68 percent of the net sown area of the wholly irrigated holdings. This proportion declined with the increase in the size of the holdings except large holdings. In case of partly irrigated holdings maximum 58.36 percent of the holdings were marginal holdings sharing 17.04 percent of the total net sown area and 17.75 percent of the total net area irrigated. In case of wholly unirrigated holdings 77.69 percent of the holdings were marginal holdings accounting for 36.33 percent of the net sown area of the total.

4.14 SOURCEWISE CLASSIFICATION OF IRRIGATED HOLDINGS

4.14.1 The following table presents the distribution of number of holdings and net area irrigated by source of irrigation.

TABLE 4.18

SOURCEWISE CLASSIFICATION OF IRRIGATION HOLDINGS AND AREA IRRIGATED IN 2010-11

Sl. No.	Source of irrigation	No. of holdings	Net area irrigated (ha.)
1	**2**	**3**	**4**
1	Canals	5941	5745
2	Tanks	1037	1163
3	Wells	7205	7002
4	Tube Wells	191	284
5	Other Sources	27489	21376
All Sources		**41863***	**35570**

* *Total number of holdings include those irrigated from more than one source.*

4.14.2 There were 41863 holdings receiving irrigation including those irrigated from more than one source. Of these, 27489 holdings which were irrigated by other sources had net irrigated area of 21376 hectares. The second important source of irrigation was well, which irrigated 7002 hectares in 7205 holdings. The canal provided irrigation to 5941 holdings with irrigated area of 5745 hectares 1037 holdings corresponding to 1163 hectares were irrigated by means of tanks.

4.14.3 As may be seen in table 4.19 below, out of the total net irrigated area of 35570 hectares, the share of marginal holdings was the highest i.e. 11443 hectares followed by small holdings with 7244 hectares. While the large holdings had a share of 6481 hectares, the medium and semi medium holdings accounted for 4704 hectares and 5698 hectares of

net irrigated area respectively. The table below gives irrigated area by source for all size classes for the year 2010-11.

TABLE 4.19

IRRIGATED AREA BY SOURCES OF IRRIGATION AND MAJOR SIZE CLASSES OF HOLDINGS IN 2010-11

Sl. No.	Major size classes	Area irrigated by					
		Canals	Tanks	Wells	Tube wells	Other sources	Total
1	2	3	4	5	6	7	8
1	Marginal	1799	356	1580	31	7677	11443
2	Small	1075	168	1290	63	4648	7244
3	Semi-Medium	746	133	1491	62	3266	5698
4	Medium	547	192	1441	9	2515	4704
5	Large	1578	314	1200	119	3270	6481
All holdings		**5745**	**1163**	**7002**	**284**	**21376**	**35570**

4.14.4 As detailed in above table, the area irrigated by the other sources of irrigation accounted for 21376 hectares of area in the State as per 2010-11 Census followed by Wells which accounted for an area of 7002 hectares. Canals provided irrigation facilities to 5745 hectares while tanks and tube wells irrigated an area of 1163 and 284 hectares of land.

4.15 CROPPING PATTERN

4.15.1 The cropping pattern generally indicates the level of development and the economic prosperity of the State. It deals with the nature of crops and percentage of area under each crop. A change in the cropping pattern would mean a change in the area proportionate under different crops. If the proportion of the area under high value crop increase it is likely to result in increase in the total return even if there is no increase in the yield rate or price, provided there is no simultaneous decrease in the proportionate area under equally high or higher value crops. The shift in cropping pattern could be advantageous or disadvantageous depending upon the crop adopted. The study of the cropping pattern is of paramount significance for planning of agricultural development as also to provide balanced programmes for cropping so that the people of this area get adequate food and raw material for agro-based industries.

4.15.2 The following table gives the distribution of area under different crops both irrigated and un-irrigated as per the Agricultural Census 2010-11.

TABLE 4.20

DISTRIBUTION OF AREA AND THEIR PERCENTAGE UNDER DIFFERENT CROPS

Sl. No.	Name of crop	Area in Hectares			Percentage to the total cropped		
		Irrigated	Un-irrigated	Total	Irrigated	Un-irrigated	Total
1	2	3	4	5	6	7	8
1	Paddy	12865	612	13477	36.16	1.16	15.29
2	Kharif Paddy	265	21643	21908	0.74	41.17	24.85
3	Ragi	42	29	71	0.12	0.06	0.08
4	Sugarcane	2092	139	2231	5.88	0.26	2.53
5	Arecanut	5186	229	5415	14.58	0.44	6.14
6	Other spices & condiments	460	161	621	1.29	0.31	0.70
7	Cashewnut	4920	24300	29220	13.83	46.23	33.15
8	Fruit(Total)	1921	930	2851	5.40	1.77	3.23
9	Vegetables (Total)	251	15	266	0.71	0.03	0.30
10	Coconut	7122	3715	10837	20.02	7.07	12.29
11	Total oil seeds	231	15	246	0.65	0.03	0.28
12	Total other food crops	3	0	3	0.01	0.00	0.00
13	Total other non food crops	112	635	747	0.31	1.21	0.85
14	Total Pulses	107	144	251	0.30	0.27	0.28
Total		**35577**	**52567**	***88144**	**100.00**	**100.00**	**100.00**

* *Gross Cropped Area*

4.15.3 Paddy during Kharif season had the share of 21908 ha. in the State. Among the cash crops Cashew nut, Coconut and Arecanut were also the most important crops. For the State as a whole 40.22 percent of the Gross Cropped Area was under total cereals. As regards the cash crops, Cashew nut accounted for 33.15 percent of the area followed by Coconut 12.29 percent and Arecanut 6.14 percent.

4.16 DISTRICTWISE DISTRIBUTION OF OPERATIONAL HOLDINGS BY SOCIAL GROUP.

In the table 4.21 below. Districtwise distribution of number of operational holdings & their corresponding area is presented according to different social groups & operational holdings.

TABLE 4.21

DISTRICTWISE DISTRIBUTION OF OPERATIONAL HOLDING BY SOCIAL GROUP AND SIZE CLASSES IN 2010-11

(Area in ha.)

Name of the District	Institutional		Scheduled Caste		Scheduled Tribe		Others		Total	
	No. of holdings	Area	No. of holding	Area	No. of holdings	Area	No. of holdings	Area	No. of holdings	Area
1	*2*	*3*	*4*	*5*	*6*	*7*	*8*	*9*	*10*	*11*
North Goa	69	1425	397	312	5486	3924	39939	45714	45891	51375
South Goa	56	626	83	52	15580	20151	16410	16790	32129	37619
Goa State Total	**125**	**2051**	**480**	**364**	**21066**	**24075**	**56349**	**62504**	**78020**	**88994**

TABLE 4.22

DISTRIBUTION OF OPERATIONAL HOLDINGS BY SOCIAL GROUP AND SIZE CLASSES IN 2010-11

Social Group	No. of holdings & area operated	Less than 1.0 ha. (Marginal)	1.0 to 2.0 ha. (Small)	2.0 to 4.0 ha. (Semi Medium)	4.0 to 10.0 ha. (Medium)	10.0 & above (Large)	Total
1	2	3	4	5	6	7	8
Institutional	No.	43	15	20	22	25	125
		(34.40)	(12.00)	(16.00)	(17.60)	(20.00)	(100.00)
	Area	20	25	57	146	1803	2051
		(0.97)	(1.22)	(2.83)	(7.12)	(87.86)	(100.00)
Scheduled Caste	No.	373	86	16	5	0	480
		(77.71)	(17.92)	(3.33)	(1.04)	(0.00)	(100.00)
	Area	136	153	43	32	0	364
		(37.36)	(42.03)	(11.82)	(8.79)	(0.00)	(100.00)
Scheduled Tribe	No.	14851	3412	2194	575	34	21066
		(70.50)	(16.20)	(10.41)	(2.73)	(0.16)	(100.00)
	Area	7330	6270	6400	3449	626	24075
		(30.45)	(26.05)	(26.58)	(14.32)	(2.60)	(100.00)
Others	No.	44633	6304	3477	1408	527	56349
		(79.21)	(11.19)	(6.17)	(2.50)	(0.93)	(100.00)
	Area	20617	11144	10270	8750	11723	62504
		(32.99)	(17.83)	(16.43)	(14.00)	(18.75)	(100.00)
Total	No.	59900	9817	5707	2010	586	78020
		(76.78)	(12.58)	(7.31)	(2.58)	(0.75)	(100.00)
	Area	28103	17591	16770	12378	14152	88994
		(31.58)	(19.77)	(18.84)	(13.91)	(15.90)	(100.00)

Note: Figures in brackets indicate percentages.

- Total may not tally exactly due to rounding off.

TABLE 4.23

TALUKAWISE DISTRICBUTION OF OPERATIONAL HOLDINGS BY SOCIAL GROUP AND SIZE CLASSES IN 2010-11

(Area in ha.)

Name of the Taluka	Institutional		Scheduled Caste		Scheduled Tribe		Others		Total	
	No. of holdings	Area	No. of holdings	Area	No. of holdings	Area	No. of holdings	Area	No. of holdings	Area
1	*2*	*3*	*4*	*5*	*6*	*7*	*8*	*9*	*10*	*11*
Tiswadi	23	89	0	0	975	741	2075	2411	3073	3241
Bardez	2	1	33	16	203	112	7819	7269	8057	7398
Pernem	10	4	89	41	0	0	9359	7799	9458	7844
Bicholim	13	40	100	69	229	159	7765	8530	8107	8797
Sattari	10	1272	146	171	770	817	7176	12981	8102	15242
Ponda	11	19	29	15	3309	2095	5745	6724	9094	8853
North Goa (District)	**69**	**1425**	**397**	**312**	**5486**	**3924**	**39939**	**45714**	**45891**	**51375**
Sanguem	13	247	40	28	3236	3443	3119	3665	6408	7383
Salcete	13	44	3	6	3254	2168	6801	4456	10071	6674
Quepem	8	102	8	9	4027	5011	2813	2746	6856	7868
Mormugao	8	17	0	0	88	30	801	706	897	753
Canacona	10	23	30	9	4516	8771	1501	2431	6057	11234
Dharbandora	4	192	2	0.40	459	728	1375	2786	1840	3707
South Goa (District)	**56**	**626**	**83**	**52**	**15580**	**20151**	**16410**	**16790**	**32129**	**37619**
Goa Sate	**125**	**2051**	**480**	**364**	**21066**	**24075**	**56349**	**62504**	**78020**	**88994**

***Total may not tally exactly due to rounding off.**

The above tables shows that out of 78020 holdings & 88994 ha. area operated in the state 125 (0.16 percent) were Institutional holding with operated area of 2051 ha. (2.30 percent), 480 (0.62 percent) holdings were scheduled Caste holding operated area of 364 ha. (0.41 percent), 21066 (27.00 percent) holdings were ST and were operated area of 24075 ha. (27.05 percent) and 56349 holdings (72.22 percent) were others operated area of 62504 ha. (70.23 percent).

CHAPTER V

SUMMARY OF FINDINGS AND LIMITATIONS

5.1 Summary of findings

5.1.1 The total number of operational holdings in Goa according to the current agricultural census was estimated to 78020 with the total operated area of 88994 hectares.

5.1.2 The total number of holdings in the State in 2010-11 agricultural census has shown a increase of 25199 holdings, and increase of 28252 hectares when compared with 2005-06 agricultural census figures i.e. 52821 holdings and 60742 hectares.

5.2 Size class with distribution of operational holdings and their area

5.2.1 The size class below 0.5 hectares count for the highest number of holdings i.e. 42414 which calculates to 54.36 percent of total holding, while the holdings in other size classes were 35606 and accounted for 45.64 %.

5.2.2 In case of operated areas the size class '1.00 – 2.00' hectares had maximum operated areas i.e. 17591 hectares which is 19.77 percent of total operated area followed by below 0.05 ha size class with 15.85 % of the total operated area, the minimum operated area 2930 hectares was observed in the size class 7.5 – 10.0 hectares which is 3.29 percent of the total operated area in Goa.

5.3 Average size class of operational holdings

The average size class of operational holding in Goa in all categories is 1.14 hectares.

5.3.1 Net area sown is 87214 hectares which is 98 percent of total operated area.

The area under current fallow is 305 hectares which is 0.34 percent of total operated area.

5.3.2 Operational holdings receiving irrigation

There were 41863 operational holdings under irrigation including those irrigated from more than one source. The net irrigated area was 35570 hectares which was 39.97 percent of total area of the holdings in Goa.

5.3.3 Irrigated and un-irrigated area under different crops

The area under irrigation formed 39.97 percent of the State holdings. The irrigated area under Paddy formed 36.90 percent of the State irrigated area. The un-irrigated area under Paddy formed 42.33 percent of the total un-irrigated area.

Intensity of cropping in Goa was 101.07 percent as per 2011 Agricultural Census.

5.3.4 District wise analysis

There are two districts in Goa viz. North Goa District and South Goa District. The number and area of the operational holding in these two districts were as follows.

North Goa District had 45891 holdings with 51375 hectares of operational area. This account for 57.73 percent of the operated area in Goa.

South Goa District had 32129 holdings with 37619 hectares of operational area. which is 42.27 percent of the operated area in Goa.

5.4 Limitations:

For Agricultural Census for 2010-11, the data has been collected through the field inquiries from all the operational holdings from the selected villages. The 50 percent of the villages/wards in each Taluka were selected by using 'Simple Random Sampling Method' and in each selected village/ward, cent percent of the operational holdings were covered during the Census. The data has been collected through oral inquiry methods. Since there is no primary system in Goa, the data has its own limitations, hence to get quality of data; primary reporting system is needed in the State.

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